

# Risk Management

Lecture # 40, 41, 42,

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SE1001 -Intro. to Software Engineering



# Today's Outline

- **Risk Definition**
- **Risk Management**
- **Risk identification**
- **Risk projection (estimation)**
- **Risk mitigation, monitoring, and management**

# What is Risk?

- Risk is an uncertainty.
- We don't know whether a particular event will occur or no but if it does has a negative impact on a project.
- An example would be that team is working on a project and the developer walks out of project and other person is recruited in his place and he doesn't work on the same platform and converts it into the platform he is comfortable with. Now the project has to yield the same result in the same time span. Whether they will be able to complete the project on time. That is the risk of schedule .

# Conceptual Definition of Risk

- A risk is a potential problem – it might happen and it might not
- Conceptual definition of risk
  - Risk concerns future happenings
  - Risk involves change in mind, opinion, actions, places, etc.
  - Risk involves choice and the uncertainty that choice entails
- Risk provides an opportunity to develop the project better.
- Risk exposure= Size (loss)\* probability of (loss)
- There is a difference between a Problem and Risk
  - Problem is some event which has already occurred but risk is something that is unpredictable.
- Two characteristics of risk
  - Uncertainty – the risk may or may not happen, that is, there are no 100% risks (those, instead, are called constraints)
  - Loss – the risk becomes a reality and unwanted consequences or losses occur
  - The probability of the risk might be assessed as very low (<10%), low (10–25%), moderate (25–50%), high (50–75%), or very high (>75%).

# Assessing Risk Impact

- Three factors affect the consequences that are likely if a risk does occur
  - Its nature – This indicates the problems that are likely if the risk occurs
  - Its scope – This combines the severity of the risk (how serious was it) with its overall distribution (how much was affected)
  - Its timing – This considers when and for how long the impact will be felt
- The overall risk exposure formula is  $RE = P \times C$ 
  - P = the probability of occurrence for a risk
  - C = the cost to the project should the risk actually occur
- Example
  - P = 80% probability that 18 of 60 software components will have to be developed
  - C = Total cost of developing 18 components is \$25,000
  - $RE = .80 \times \$25,000 = \$20,000$

# Example1: Risk Impact Assessment

- A food delivery startup is building its rider-tracking module. There is a risk that **GPS inaccuracies** may occur due to weak signals in urban areas.
- Probability of GPS glitch: **0.2**
- If it occurs, delivery delays will cause customer refunds.
- Estimated refunds range from **25,000 to 55,000 PKR**, with the most likely value being **40,000 PKR**.
- **Question:**
- Using the most likely impact, calculate the **Risk Exposure** for the GPS inaccuracy risk.

# Example 2: Risk Impact Assessment

- A mobile app project is expected to face a risk of **delayed API integration** from a third-party vendor.
- The estimated probability of delay is **0.35**.
- If this delay occurs, the project will slip by **15 days**.
- The development team costs the company **12,000 PKR per day**.
- **Question:**
- Determine the **Risk Exposure** and briefly explain what this value means for the project manager.

# Risk Categorization

- **Project risks**
  - They threaten the project plan
  - If they become real, it is likely that the project schedule will slip and that costs will increase
- **Technical risks**
  - They threaten the quality and timeliness of the software to be produced
  - If they become real, implementation may become difficult or impossible
- **Business risks**
  - They threaten the viability of the software to be built
  - If they become real, they jeopardize the project or the product

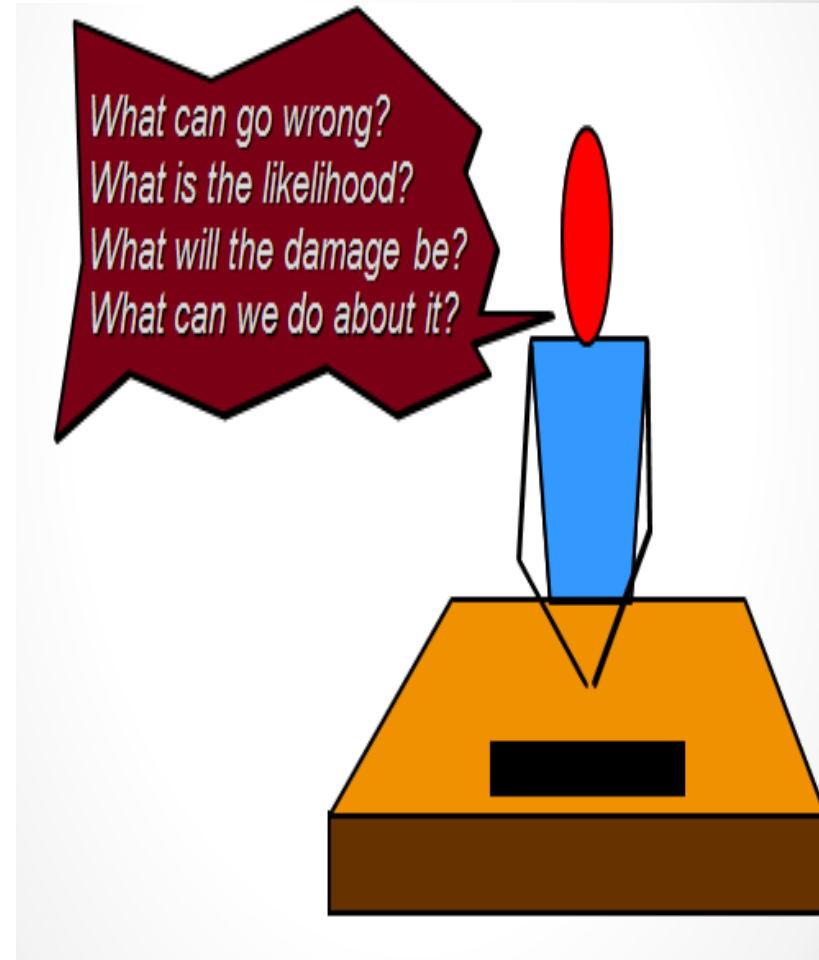


# Risk Categorization – (continued)

- Sub-categories of Business risks
  - **Market risk** – building an excellent product or system that no one really wants
  - **Strategic risk** – building a product that no longer fits into the overall business strategy for the company
  - **Sales risk** – building a product that the sales force doesn't understand how to sell
  - **Management risk** – losing the support of senior management due to a change in focus or a change in people
  - **Budget risk** – losing budgetary or personnel commitment

# Risk Management

- The Risks we encounter in a project should be resolved so that we are able to deliver the desired project to the customer.
- The project should be managed in such a way that the risks don't affect the project in a big way.
- The art of managing of the risks effectively so that the WIN-WIN situation and friendly relationship is established between the team and the customer is called Risk Management.
- By using various paradigms, principles we can manage the risks.



# Reactive vs. Proactive Risk Management Strategies

- **Reactive risk strategies**
  - "Don't worry, I'll think of something"
  - The majority of software teams and managers rely on this approach
  - Nothing is done about risks until something goes wrong
  - project team reacts to risks when they occur
  - fix on failure—resources are found and applied when the risk strikes
  - Crisis management is the choice of management techniques

# Reactive vs. Proactive Risk Management Strategies

- **Proactive risk strategies**
- **formal risk analysis is performed**
- **organization corrects the root causes of risk**
  - **Steps for risk management are followed**
  - **Primary objective is to avoid risk and to have a contingency plan in place to handle unavoidable risks in a controlled and effective manner**

# Steps for Proactive Risk Management

- 1) Identify possible risks; recognize what can go wrong
- 2) Analyze each risk to estimate the probability that it will occur and the impact (i.e., damage) that it will do if it does occur
- 3) Rank the risks by probability and impact
  - Impact may be negligible, marginal, critical, and catastrophic
- 4) Develop a contingency plan to manage those risks having high probability and high impact



# Contents of a Risk Table

- A risk table provides a project manager with a simple technique for risk projection
- It consists of five columns
  - Risk Summary – short description of the risk
  - Risk Category – one of seven risk categories (slide 12)
  - Probability – estimation of risk occurrence based on group input
  - Impact – (1) catastrophic (2) critical (3) marginal (4) negligible
  - RMMM – Pointer to a paragraph in the Risk Mitigation, Monitoring, and Management Plan

| Risk Summary | Risk Category | Probability | Impact (1-4) | RMMM |
|--------------|---------------|-------------|--------------|------|
|              |               |             |              |      |
|              |               |             |              |      |
|              |               |             |              |      |

# Developing a Risk Table

- List all risks in the first column (by way of the help of the risk item checklists)
- Mark the category of each risk
- Estimate the probability of each risk occurring
- Assess the impact of each risk based on an averaging of the four risk components to determine an overall impact value
- Sort the rows by probability and impact in descending order
- Draw a horizontal cutoff line in the table that indicates the risks that will be given further attention

# Impact Level

| Impact Level    | Cost Impact           |
|-----------------|-----------------------|
| 1. Catastrophic | Budget overrun >20%   |
| 2. Critical     | Budget overrun 10-20% |
| 3. Marginal     | Budget overrun 5-10%  |
| 4. Negligible   | Budget overrun <5%    |



# Risk Mitigation, Monitoring, and Management Background

- An effective strategy for dealing with risk must consider three issues
  - Risk mitigation (i.e., avoidance)
  - Risk monitoring
  - Risk management and contingency planning
- Risk mitigation (avoidance) is the primary strategy and is achieved through a plan
  - Example: Risk of high staff turnover

# RMMM Table Example

| Risk Summary                               | Risk Category | Probability | Impact       | RMMM Plan                                                                                                                                                                        |
|--------------------------------------------|---------------|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developer may leave during project         | Project       | 0.3 (30%)   | Critical     | Mitigation: Keep backup resource, share knowledge. Monitoring: Weekly check-ins on workload & morale. Management: Reassign tasks and increase team support if developer resigns. |
| API from third-party vendor may be delayed | Technical     | 0.4 (40%)   | Marginal     | Mitigation: Establish communication early; request sandbox. Monitoring: Track vendor updates. Management: Use mock API if delay occurs.                                          |
| Requirement changes by client              | Project       | 0.25 (25%)  | Critical     | Mitigation: Freeze requirements; use change-control process. Monitoring: Weekly client meetings. Management: Adjust timeline & update scope documents.                           |
| Server downtime due to high load           | Technical     | 0.2 (20%)   | Catastrophic | Mitigation: Add load balancing, stress testing. Monitoring: Use performance monitoring tools. Management: Switch to backup server or scale cloud resources.                      |

# RMMM Table Example

| Risk Summary                                    | Risk Category | Probability | Impact     | RMMM Plan                                                                                                                                                                                                                      |
|-------------------------------------------------|---------------|-------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minor UI defects discovered during testing      | Technical     | 0.5 (50%)   | Negligible | Mitigation: Follow UI guidelines.<br>Monitoring: Frequent design reviews.<br>Management: Fix defects in next sprint.                                                                                                           |
| Regulatory or legal changes impose new costs    | Business      | 0.10        | Critical   | Mitigation: Stay informed of regulatory trends, consult legal advisors.<br>Monitoring: Regular compliance reviews and legal audits.<br>Management: Redesign features, budget for compliance, or delay launch.                  |
| Competitor launches a similar product earlier   | Business      | 0.25        | Critical   | Mitigation: Accelerate development, secure unique features, protect IP.<br>Monitoring: Monitor competitor announcements and market news.<br>Management: Increase marketing spend, highlight differentiators.                   |
| Market demand may decrease after product launch | Business      | 0.30        | Critical   | Mitigation: Conduct market research, run pilot sales, prepare promotional campaigns.<br>Monitoring: Track customer feedback, sales metrics monthly.<br>Management: Adjust pricing or pivot product features if demand weakens. |



That is all